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```
%ME103 CUSTOM LAB DATA ANALYSIS
%compare thrust, power, and noise levels between blade configs
```

Load Data

```
BladeRPM = [1855 2636 3288 3779];
BladedB = [66 71.5 72.5 78.3];
BladeThr = [3.563 7.541 15.472 21.735];
BladePow = [2103.006 3034.535 4049.497 5118.774];
BladeEff = BladeThr./BladePow;

TorRPM = [1115 1500 1805 2087];
TordB = [65.5 68.3 77 74.7];
TorThr = [4.068 7.147 11.160 15.230];
TorPow = [2188.952 3332.252 4675.529 6250.444];
TorEff = TorThr./TorPow;
```

Fit, Extrapolate Data

```
TordBsm = smooth(TorRPM, TordB, 0.8, "rlowess");
BladedBsm = smooth(BladeRPM, BladedB, 0.8, "rlowess");

TorRPMfit = linspace(1000,4000, 10);
p = polyfit(TorRPM, TorEff, 1);
TorEfffit = polyval(p,TorRPMfit);
```

Setup Plots

```
tiledlayout(2,2)

nexttile;
plot(BladeRPM, BladeEff);
hold on
plot(TorRPM, TorEff);
plot(TorRPMfit, TorEfffit);
title("Propeller RPM vs Efficiency");
xlabel("RPM");
ylabel("Efficiency (Thrust/Power)");
legend("3-blade", "Toroidal", "ToroidalFit");
hold off

nexttile;
% plot(BladeEff, BladedB);
```

```

hold on
plot(BladeEff, BladedBsm);
% plot(TorEff, TordB);
plot(TorEff, TordBsm);
title("Propeller Efficiency vs Noise Output");
xlabel("Efficiency");
ylabel("Smoothed Noise (dBA)");
legend("3-blade", "Toroidal");

nexttile;
plot(BladeRPM, BladedBsm);
hold on
plot(TorRPM, TordBsm);
title("Propeller RPM vs Noise Output");
xlabel("RPM");
ylabel("Smoothed Noise (dBA)");
legend("3-blade", "Toroidal");
hold off

nexttile;
plot(BladeThr, BladedBsm);
hold on
plot(TorThr, TordBsm);
title("Propeller Thrust vs Noise Output");
xlabel("Thrust");
ylabel("Smoothed Noise (dBA)");
legend("3-blade", "Toroidal");
hold off

```

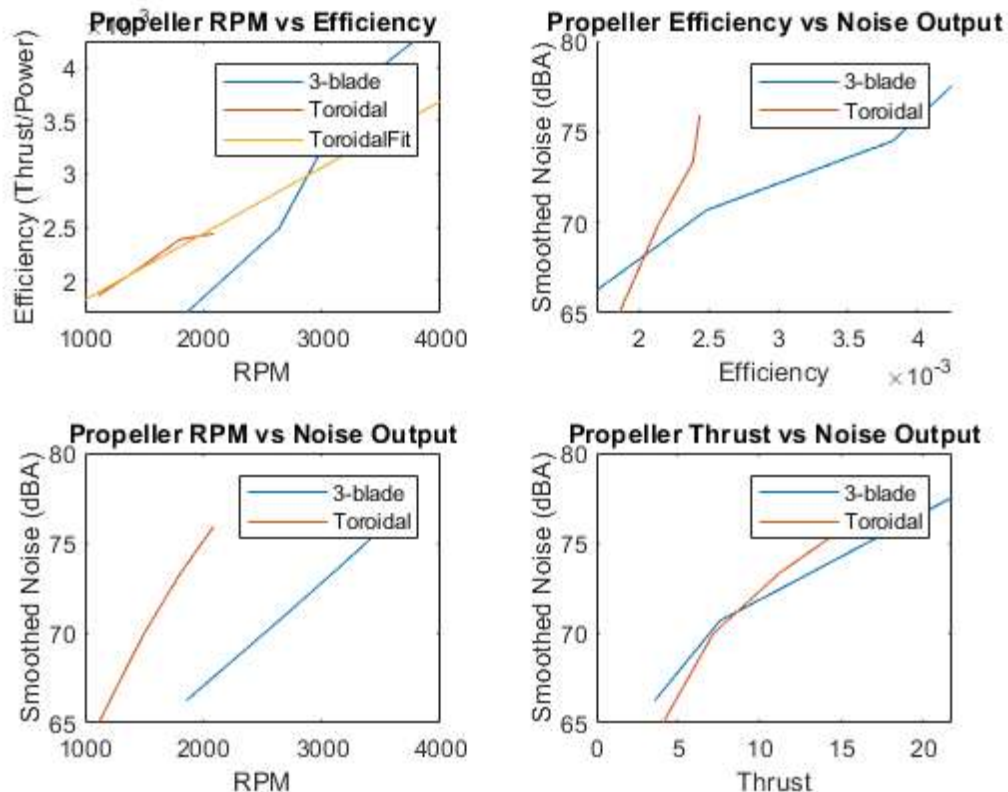
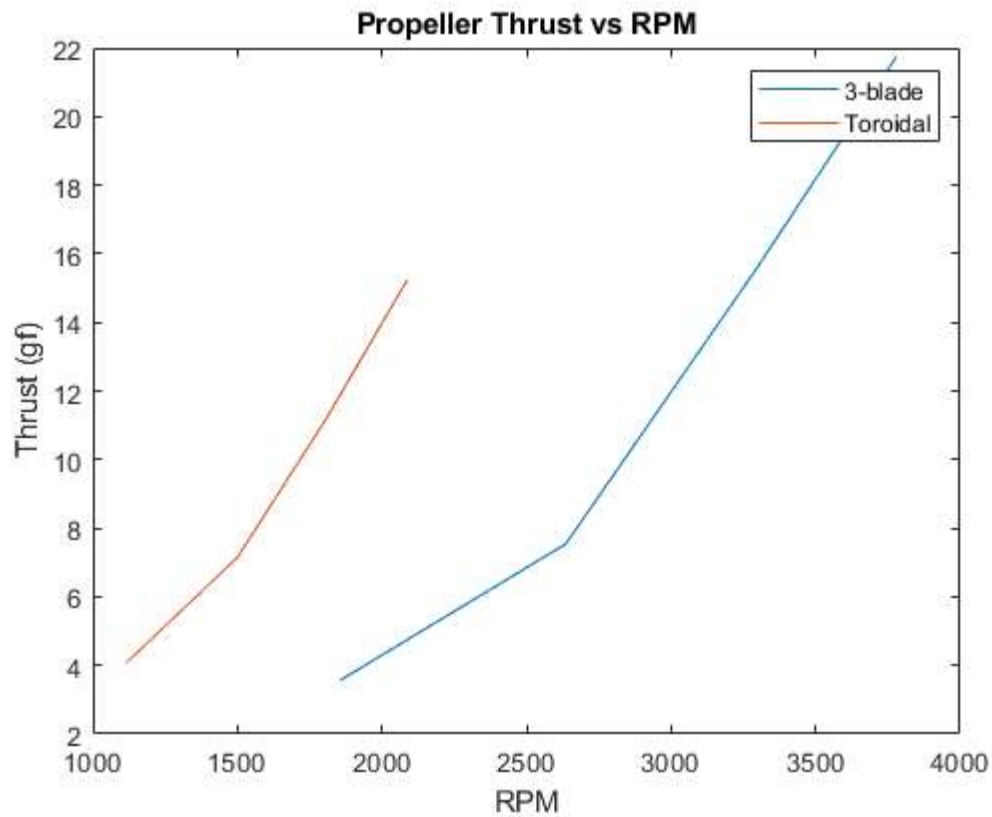
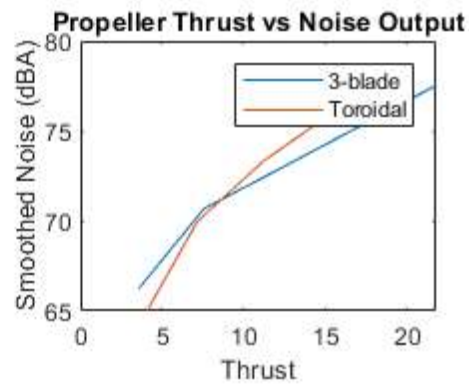
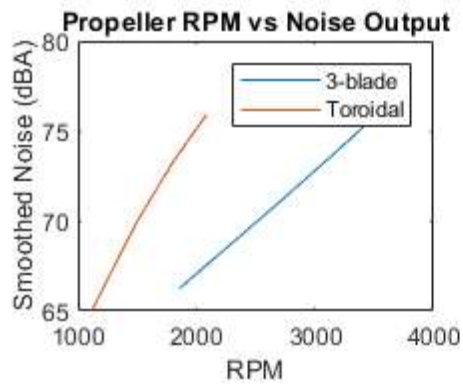
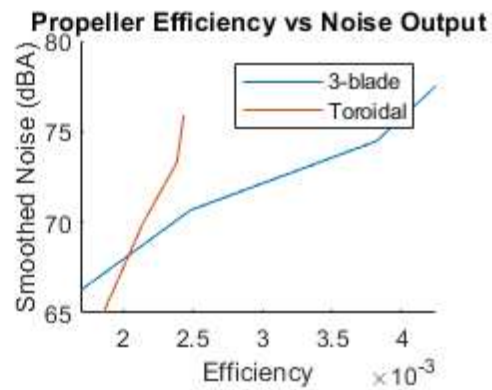
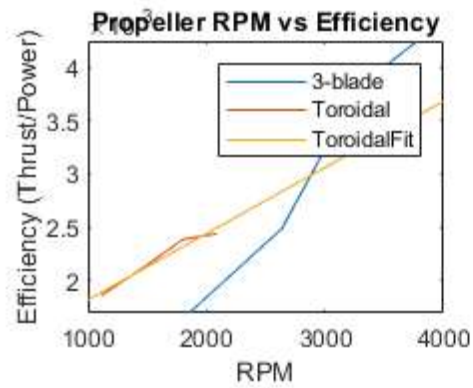


Figure Plots

```
figure(2)
plot(BladeRPM, BladeThr);
hold on
plot(TorRPM, TorThr);
title("Propeller Thrust vs RPM");
xlabel("RPM");
ylabel("Thrust (gf)");
legend("3-blade", "Toroidal");
hold off
```



Theory, Assuming $Re > 80000$

%equations and coefficients obtained from He et. al., doi 10.1109/ACCESS.2017.2725303

```
RPM = linspace(1000, 4000, 50);
```

```

BladeThrPred = zeros(1, length(RPM));
BladeEffPred = zeros(1, length(RPM));
TorThrPred = zeros(1, length(RPM));
TorEffPred = zeros(1, length(RPM));

for i = 1:length(RPM)
[BladeThrPred(i), BladeEffPred(i)] = efficiency(RPM(i), 15);
[TorThrPred(i), TorEffPred(i)] = efficiency(RPM(i), 30);
end

figure(3)
plot(RPM, BladeThrPred);
% hold on
% plot(RPM, TorThrPred);
%plot(RPM, BladeEffPred);
title("RPM vs Theoretical Thrust (15deg AoA)");
xlabel("RPM");
ylabel("Thrust (gf)");
hold off

function [T, Eff] = efficiency(RPM, AOA)
rho = 1.29; %kg/m3
n = RPM*0.1047; %rad/s
diam = 0.102; %meters
b = .021; %average chord length, in to m
V = pi*n*diam; %tip velocity
vinf = 0.001; %m/s
vt = 0.06; %tangential velocity

if AOA == 15
    Cd = .125;
    Cl = 2.1;
else %AOA = 30. Unreliable.
    Cd = 1.0;
    Cl = 2.0;
end

a = 1; %unity of va/vinf
gamma = a*tan(Cd/Cl); %ratio of drag to lift
phi = pi/2-vt; %angle of air velocity to sweep direction

T = 1/2*rho*V^2*3*(diam/2)*Cl*b*(1+a)^2/(sin(phi)^2*cos(gamma))*cos(gamma + phi); %Newtons
T = T*101.97; %gf
M = 1/2*rho*V^2*3*(diam^2/8)*Cl*b*(1+a)^2/(sin(phi)^2*cos(gamma))*sin(gamma + phi);

% CT = T/(rho*n^2*D^4);
% CP = (2*pi*n*M)/(rho*n^3*D^5);

lambda = 0.8;
Eff = T/M*vinf/(2*pi); %CT/CP*lambda;

end

```

